

Title: AI-Based Financial News Sentiment Analysis for Stock Market Prediction

Abstract

Financial news plays a significant role in influencing stock market trends, making sentiment analysis an essential tool for investors and analysts. Traditional methods of analyzing financial news are often time-consuming and may fail to capture market sentiment efficiently. This project proposes an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that automatically collects financial news articles from online sources, analyzes the sentiment expressed in the news, and predicts the possible movement of stock prices.

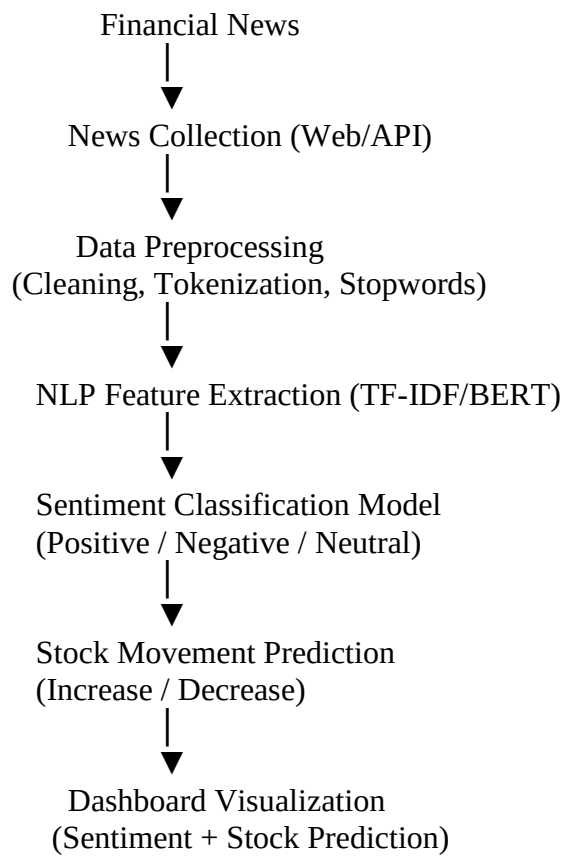
The proposed system uses Natural Language Processing (NLP) techniques along with Machine Learning models to preprocess financial news, extract meaningful features, and classify each news article into Positive, Negative, or Neutral sentiment categories. Based on the detected sentiment, the system predicts whether the corresponding stock price is likely to Increase or Decrease. The processed information and predictions are presented through an interactive dashboard, enabling users to easily understand market trends and make informed investment decisions.

The performance of the sentiment analysis model is evaluated using standard metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The resulting system aims to provide timely and reliable insights from financial news, assisting traders, investors, and researchers in understanding market sentiment and supporting better financial decision-making.

Datasets

Dataset	URL link
Financial Sentiment Analysis	https://www.kaggle.com/datasets/williamtage/trace-acl18?utm_source=chatgpt.com
Financial PhraseBank	https://www.kaggle.com/datasets/ankurzing/sentiment-analysis-for-financial-news?utm_source=chatgpt.com
<u>TRACE ACL18</u>	https://www.kaggle.com/datasets/sbhatti/financial-sentiment-analysis?utm_source=chatgpt.com

Approach



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